

TOC:-

→ Alphabet (Σ):- finite, non empty set of symbols, atomic

string:- finite sequence of symbols over the given alphabet. $\{a, b\}, \{x\}$

Deterministic finite Automata (DFA):-

$DFA = (Q, \Sigma, q_0, f, \delta)$

$Q \Rightarrow$ set of finite states

$\Sigma \Rightarrow$ input alphabet

$q_0 \Rightarrow$ initial state

$Q \times \Sigma \rightarrow Q$

$f \Rightarrow$ set of final state

$\delta \Rightarrow$ transition function

Methods to check DFA:-

Method-1:-

→ if dependency is there in a and b then DFA is not possible.

① $L = \{a^n b^n \mid n \geq 0\} \rightarrow$ DFA not possible

Method-2:-

→ if common difference is exist then $\Rightarrow$ DFA possible

→ if no common difference $\Rightarrow$ DFA not possible.

$L = \{2^n \mid n \geq 1\} = \{2, 4, 8, 16, \dots\}$

No common difference so DFA not possible.

Note:- ① every finite language is regular

② Infinite + composition $\Rightarrow$ not regular

③ Infinite + no composition $\Rightarrow$ regular

Prefix & suffix:-

Ex:- abc

① Prefix:- $\begin{matrix} abc \\ \epsilon \\ a \\ ab \\ abc \end{matrix} = \epsilon, a, ab, abc$

② suffix:- $\begin{matrix} abc \\ c \\ bc \\ abc \end{matrix} = \epsilon, c, bc, abc$

Note:- ① Number of prefix and suffix are Always same and equal to $(n+1)$

② Number of maximum substring having n length string:-

Range $\Rightarrow (n+1, \frac{n(n+1)}{2} + 1)$

③ if $|w| = n \Rightarrow |wR| = n$

④ if $|w_1| = m$ and $|w_2| = n \Rightarrow |w_1 w_2| = |w_2 w_1| = m+n$

Language:-

Reg $\Rightarrow$ CFL $\Rightarrow$ CSL $\Rightarrow$ REL

Automata:-

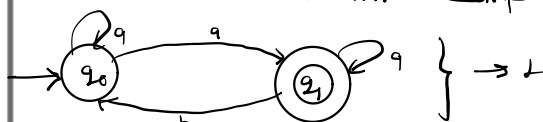
FA < PDA < LBA < TM (Expressive power)

Grammar:-

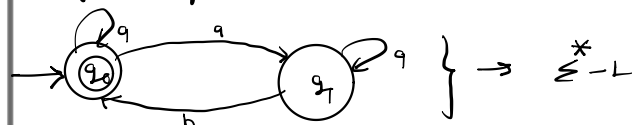
RL $\Rightarrow$ CFL $\Rightarrow$ CSL

Complement of DFA:- ($L = \Sigma^* - L$)

→ By interchanging final & non final states we can convert into complement of DFA



Complement:-



the DFA which is accepted by (A) will be rejected by complement (B) vice versa

Note:-

to find complement of NFA, first convert NFA to DFA

→ don't simply change final, non final like DFA.

NFA:-

→ every DFA is NFA but every NFA need not be DFA

→ every DFA can be converted in NFA and NFA can be converted to DFA or minimal DFA.

$Q \times \Sigma \rightarrow 2^Q$